

1.

a. $3^x = 6 \pmod{7}$

$$\begin{aligned} N &= (\sqrt{p} - 1) + 1 \\ &= (\sqrt{7} - 1) + 1 \\ &= 4 \end{aligned}$$

Baby step

a^j where 4 for $0 \leq j < 4$

$$3^0 = 1 \pmod{7}$$

$$3^1 = 3 \pmod{7}$$

$$3^2 = 2 \pmod{7}$$

$$3^3 = 6 \pmod{7}$$

Giant step

$$Ba^{-NK}$$

$$K = 0, 6 \times (3^{-4 \cdot 0}) = 6$$

$$K = 1, 6 \times (3^{-4 \cdot 1}) = 5$$

$$K = 2, 6 \times (3^{-4 \cdot 2}) = 3$$

$$K = 3, 6 \times (3^{-4 \cdot 3}) = 6$$

We have two matches

First match

$$3^1 = 6 \times 3^{-8}$$

$$3 = 3$$

$$3^1 / 3^{-8} = 6$$

$$3^1 \times 3^8 = 6$$

$$3^9 = 6 \pmod{7}$$

Second match

$$3^3 = 6 \times 3^{-12}$$

$$6 = 6$$

$$3^3 / 3^{-12} = 6$$

$$3^3 \times 3^{12} = 6$$

$$3^{15} = 6 \pmod{7}$$

The choice for X seems to be 9 and 15

b.

$q = 4$ and $a = 9$

private keys $P1 = 3$ and $P2 = 6$

$$9^3 = 4 \pmod{29}$$

$$9^6 = 16 \pmod{29}$$

$P1$ takes $P2$'s public value 16 and uses the exponent to get:

$$16^3 = 7 \pmod{29}$$

$P2$ takes $P1$'s public value 4 and use it with his exponent to get:

$$4^6 = 7 \pmod{29}$$

They can now use $k = 7$ as a shared key

c.

First, I'm choosing $m=10$, shared key, $k = 7$ and $p = 29$

Encryption

$$\begin{aligned} C &= k * m \pmod{p} \\ &= 7 * 10 \pmod{29} \\ &= 12 \end{aligned}$$

Decryption

$$C = k * m \pmod{p}$$

$$\begin{aligned} m &= c * k^{-1} \pmod{p} \\ &= (12 * 7^{-1}) \pmod{29} \\ &= 12 * 25 \pmod{29} \\ &= 10 \end{aligned}$$

The message is decrypted

2.

a.

$$Y = (4x + 3) \bmod 7$$

$$X = 0, 1, 2, 3, 4, 5, 6$$

$$\begin{aligned} Y &= (4 * 0 + 3) \bmod 7 \\ &= 3 \bmod 7 \\ &= 3 \rightarrow \text{the secret} \end{aligned}$$

$$\begin{aligned} Y &= (4 * 1 + 3) \bmod 7 \\ &= 7 \bmod 7 \\ &= 0 \end{aligned}$$

$$\begin{aligned} Y &= (4 * 2 + 3) \bmod 7 \\ &= 11 \bmod 7 \\ &= 4 \end{aligned}$$

$$\begin{aligned} Y &= (4 * 3 + 3) \bmod 7 \\ &= 15 \bmod 7 \\ &= 1 \end{aligned}$$

$$\begin{aligned} Y &= (4 * 4 + 3) \bmod 7 \\ &= 19 \bmod 7 \\ &= 5 \end{aligned}$$

$$\begin{aligned} Y &= (4 * 5 + 3) \bmod 7 \\ &= 23 \bmod 7 \\ &= 2 \end{aligned}$$

$$\begin{aligned} Y &= (4 * 6 + 3) \bmod 7 \\ &= 27 \bmod 7 \\ &= 6 \end{aligned}$$

(0, 3), (1, 0), (2, 4), (3, 1), (4, 5), (5, 2), (6, 6)

b.

Let's take (1, 0) and (2, 4)

$$Y = (ax + b) \bmod 7$$

$$0 = (a + b) \bmod 7 \rightarrow \mathbf{1}$$

$$Y = (ax + b) \bmod 7$$

$$4 = (2a + b) \bmod 7 \rightarrow \mathbf{2}$$

$$\mathbf{2 - 1}$$

$$(2a + b) - (a + b) = 4 - 0$$

$$2a + b - a - b = 4$$

$$a = 4$$

a = 4 substitute equation 1

$$a + b = 0$$

$$4 + b = 0$$

$$b = -4$$

$$b = -4 \bmod 7$$

$$= 3$$

The secret can be constructed using any two points.

c.

$$Y = (x^2 + 5x + 3) \bmod 11$$

$$X = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (0^2 + 5 \cdot 0 + 3) \bmod 11 \\ &= 3 \bmod 11 \\ &= 3 \end{aligned}$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (1^2 + 5 \cdot 1 + 3) \bmod 11 \\ &= 9 \bmod 11 \\ &= 9 \end{aligned}$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (2^2 + 5 \cdot 2 + 3) \bmod 11 \\ &= 17 \bmod 11 \\ &= 6 \end{aligned}$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (3^2 + 5 \cdot 3 + 3) \bmod 11 \\ &= 27 \bmod 11 \\ &= 5 \end{aligned}$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (4^2 + 5 \cdot 4 + 3) \bmod 11 \\ &= 39 \bmod 11 \\ &= 6 \end{aligned}$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (5^2 + 5 \cdot 5 + 3) \bmod 11 \\ &= 53 \bmod 11 \\ &= 9 \end{aligned}$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (6^2 + 5 \cdot 6 + 3) \bmod 11 \\ &= 69 \bmod 11 \\ &= 3 \end{aligned}$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (7^2 + 5 \cdot 7 + 3) \bmod 11 \\ &= 87 \bmod 11 \\ &= 10 \end{aligned}$$

$$\begin{aligned} Y &= (x^2 + 5x + 3) \bmod 11 \\ &= (8^2 + 5 \cdot 8 + 3) \bmod 11 \\ &= 107 \bmod 11 \\ &= 8 \end{aligned}$$

$$\begin{aligned}
 Y &= (x^2 + 5x + 3) \bmod 11 \\
 &= (9^2 + 5 \cdot 9 + 3) \bmod 11 \\
 &= 129 \bmod 11 \\
 &= 8
 \end{aligned}$$

$$\begin{aligned}
 Y &= (x^2 + 5x + 3) \bmod 11 \\
 &= (10^2 + 5 \cdot 10 + 3) \bmod 11 \\
 &= 153 \bmod 11 \\
 &= 10
 \end{aligned}$$

(0, 3) (1, 9) (2, 6) (3, 5) (4, 6) (5, 9) (6, 3) (7, 10) (8, 8) (9, 8) (10, 10)

d.

Let's select (1, 9) (2, 6) (3, 5)

$$Y = (ax^2 + bx + M) \bmod 11$$

$$\begin{aligned}
 9 &= (a + b + M) \bmod 11 \rightarrow \mathbf{1} \\
 6 &= (4a + 2b + M) \bmod 11 \rightarrow \mathbf{2} \\
 5 &= (9a + 3b + M) \bmod 11 \rightarrow \mathbf{3}
 \end{aligned}$$

$$\begin{aligned}
 &\mathbf{1 \times 4} \\
 36 &= (4a + 4b + 4M) \rightarrow \mathbf{4}
 \end{aligned}$$

$$\begin{aligned}
 &\mathbf{4 - 2} \\
 (4a + 4b + 4M) - (4a + 2b + M) &= 36 - 6 \\
 4a + 4b + 4M - 4a - 2b - M &= 30 \\
 2b + 3M &= 30 \rightarrow \mathbf{5}
 \end{aligned}$$

$$\begin{aligned}
 &\mathbf{1 \times 9} \\
 81 &= (9a + 9b + 9M) \rightarrow \mathbf{6}
 \end{aligned}$$

$$\begin{aligned}
 &\mathbf{6 - 3} \\
 (9a + 9b + 9M) - (9a + 3b + M) &= 81 - 5 \\
 9a + 9b + 9M - 9a - 3b - M &= 76 \\
 6b + 8M &= 76 \rightarrow \mathbf{7}
 \end{aligned}$$

$$\begin{aligned}
 &\mathbf{5 \times 3} \\
 6b + 9M &= 90 \rightarrow \mathbf{8}
 \end{aligned}$$

$$\begin{aligned}
 &\mathbf{8 - 7} \\
 (6b + 9M) - (6b + 8M) &= 90 - 76 \\
 6b + 9M - 6b - 8M &= 14 \\
 M &= 14 \\
 M &= 14 \bmod 11 \\
 M &= 3
 \end{aligned}$$

So, yes, the key can be constructed using the 3 points

3.

a.

$$Y^2 = (x^3 + x + 1) \bmod 13$$

$$(1, 4) (4, 2)$$

$$Y^2 = x^3 + x + 1$$

$$4^2 = 1^3 + 1 + 1$$

$$16 = 1 + 1 + 1$$

$$16 = 3$$

$$16 \bmod 13 = 3 \bmod 13$$

$$3 = 3$$

$$(1, 4) (4, 2)$$

$$Y^2 = x^3 + x + 1$$

$$2^2 = 4^3 + 4 + 1$$

$$4 = 64 + 4 + 1$$

$$4 = 69$$

$$4 \bmod 13 = 69 \bmod 13$$

$$4 = 4$$

b.

$$\lambda = [(y_2 - y_1) / (x_2 - x_1)] \bmod 13$$

$$x_3 = [\lambda^2 - x_1 - x_2] \bmod 13$$

$$y_3 = [\lambda (x_1 - x_3) - y_1] \bmod 13$$

$$(1, 4) (4, 2)$$

$$\lambda = [(y_2 - y_1) / (x_2 - x_1)] \bmod 13$$

$$= [(2 - 4) / (4 - 1)] \bmod 13$$

$$= [-2/3] \bmod 13$$

$$= -2 \times 3^{-1} \bmod 13$$

$$= -2 \times 9$$

$$= -18$$

$$x_3 = [\lambda^2 - x_1 - x_2] \bmod 13$$

$$= [(-18)^2 - 1 - 4] \bmod 13$$

$$= [324 - 1 - 4] \bmod 13$$

$$= 319 \bmod 13$$

$$= 7$$

$$y_3 = [\lambda (x_1 - x_3) - y_1] \bmod 13$$

$$= [-18 (1 - 7) - 4] \bmod 13$$

$$= 104 \bmod 13$$

$$= 0$$

$$R = (7, 0)$$

Now to prove that this point lies in the curve $Y^2 = x^3 + x + 1$

$$Y^2 = x^3 + x + 1$$

$$0^2 = 7^3 + 7 + 1$$

$$0 = 351$$

$$0 \bmod 13 = 351 \bmod 13$$

$$0 = 0$$

Therefore, the point $(7, 0)$ is on the curve $Y^2 = x^3 + x + 1$